D 31828 **(Pages : 2)** **Name.....**

Reg. No.....

THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2022

Physics/Applied Physics

PHY 3B 03/APH 3B 03—ELECTRODYNAMICS—I

(2019 Admission onwards)

Time : Two Hours Maximum : 60 Marks

The symbols used in question paper have their usual meanings.

Section A (Short Answer Type)

Answer **all** *questions in two or three sentences.*
Each correct answer carries a maximum of 2 marks.

1. What do you mean by the Laplacian of a scalar ? Is it a scalar or a vector ?
2. Write down an expression for the electric field due to a volume charge. Discuss the terms involved.
3. Give Poisson's equation and explain the terms involved.
4. How is the work and energy related in electrostatics ? Is the electrostatic force conservative ?
5. Write down the expression for the torque experienced by a dipole in uniform electric field. What is the effect of the torque ?
6. Write down the relation connecting permittivity and susceptibility in linear dielectrics. Explain the terms involved. What is the susceptibility of vacuum ?
7. Check whether the following statements are true or false : (i) Magnetic forces can alter the direction in which a charged particle moves ; and (ii) Magnetic forces can speed up a charged particle.
8. Write and explain the expressions for the divergence and curl of B in magnetostatics.
9. Explain the equation of continuity.
10. What is the relation connecting magnetic flux density B and magnetic vector potential A ? What is the divergence of A ?
11. When will you say that a medium is magnetically polarized ? Discuss the magnetic polarization in diamagnets.
12. Explain the temperature-driven phase transitions in ferromagnetic materials.

(Ceiling 20)

Turn over

Section B (Paragraph/Problem Type)

Answer **all** *questions in a paragraph of about* **half a page** *to* **one page**.
Each question answer carries a maximum of 5 *marks.*

13. Prove that the curl of the gradient of a function is zero.

14. Find the electric field (magnitude and direction) at a distance z above the midpoint between two equal charges q, a distance d apart.

15. Suppose the electric field in some region is found to be $\mathrm{E} = \mathrm{kr}^3\hat{\mathrm{r}}$, in spherical polar coordinates, where k is some constant. Determine the charge density.

16. Explain why the electric field and charge density vanishes inside a conductor.

17. Compare the curl of E and D in electrostatics.

18. Give an expression for the field of a magnetized object. Explain the physical meaning of the expression.

19. Explain the hysteresis loop of ferromagnets.

(Ceiling 30)

Section C (Essay type)

Answer in about **two pages**, *any one question.*
Answer carries 10 *marks.*

20. What do you mean by the curl of a vector function ? Explain its geometrical interpretation. Explain Stokes' theorem and give its geometrical interpretation.

21. Using suitable figures, discuss the magnetostatic boundary conditions.

(1 × 10 = 10 marks)